Physiology-Guided Temporal Fusion for Cross-Subject EEG–fNIRS Brain–Computer Interfaces

Hongjie Chen and Fang Duan

Abstract*—*Objective: **To improve cross-subject EEG–fNIRS decoding by incorporating neurovascular delay priors into multimodal fusion.** ***Approach:*** **NVC-DelayFormer combines convolutional EEG and optical encoders through cross-attention with delay priors and token timestamp masks. Entropy-guided gating integrates deep attention features with spectral summaries. Evaluation used participant-isolated leave-one-subject-out validation, with separate participants for training, checkpoint selection, and testing.** ***Main results:*** **Across 29 participants and three seeds, mean accuracy was 78.22%, compared with 72.89% for standard 2D-EEGNet ($p = 1.10 \times 10^{-4}$), with primary-seed accuracy reaching 79.37%. Under trace-matched source reference initialization, recursive covariance alignment reduced relative covariance error from 45.15% after one trial to 29.38% after eight trials. In task-specific evaluations, accuracy was 73.24% for working memory and 66.32% for motor imagery.** ***Significance:*** **By embedding neurovascular timing constraints, NVC-DelayFormer achieves cross-subject generalization across cognitive and sensorimotor tasks. Operating with 63,932 parameters ($17.5\times$ smaller than representative transformers) and 2.22 ms single-trial forward execution latency on an NVIDIA RTX 4060 GPU, the compact architecture is suitable for prospective zero-calibration BCI deployment.**

***Index Terms*—Brain–computer interface, cross-subject decoding, electroencephalography, functional near-infrared spectroscopy, neurovascular coupling, temporal fusion.**

I. Introduction

BEDSIDE assistive neurotechnologies and neurorehabilitation systems require immediate, calibration-free operation to eliminate lengthy patient-specific setup routines. Hybrid brain–computer interfaces (BCIs) combining electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) improve decoding fidelity by capturing complementary electrophysiological transients and localized capillary oxygenation dynamics [1]–[3]. Translating hybrid systems to zero-calibration, cross-subject deployment faces a fundamental biophysical bottleneck: disparate temporal dynamics. Neuronal depolarization elicits spectral shifts within tens of milliseconds [4], [5], whereas neurovascular coupling (NVC) and capillary perfusion evolve over multi-second hemodynamic delays [6]–[8]. In calibration-free leave-one-subject-out (LOSO) regimes, unconstrained neural models exploit spurious temporal alignments, which impairs generalization to unseen individuals.

Conventional multimodal fusion relies on handcrafted feature concatenation with linear discriminant analysis or support vector machines [1], [9], [10]. Although Euclidean alignment (EA) mitigates inter-subject covariance shifts [11]–[13], handcrafted pipelines treat temporal windows statically and cannot resolve dynamic latency. Deep learning architectures capture spatio-temporal representations from raw biosignals [14]–[16], including compact convolutional networks [14], attention networks [17], transformer backbones [18], and optical hemoglobin networks [19], [20]. However, unconstrained cross-modal attention [21] permits acausal alignments where delayed hemodynamics attend to future neural activations ($\delta_{ij} < 0$), creating optimization artifacts that do not transfer across subjects. The hemodynamic response peak [6], [22] motivates the 3.5–6.5 s range used for the Gaussian delay prior. The accompanying token timestamp mask enforces temporal precedence between EEG keys and fNIRS queries.

To account for the temporal mismatch between EEG and fNIRS, we introduce NVC-DelayFormer for calibration-free cross-subject decoding. The principal contributions are threefold: (1) *Biophysically constrained causal cross-attention:* We couple a 2D-EEGNet backbone with a continuous optical encoder through multi-head cross-attention. Continuous Gaussian hemodynamic delay priors ($\tau_h \in [3.5, 6.5]$ s) and token timestamp causal masks ($t_{\text{nirs}} \geq t_{\text{eeg}}$) enforce directional temporal precedence between EEG and fNIRS token pairs. (2) *Uncertainty-aware entropy-guided gating:* A learned bottleneck gating network arbitrates between deep cross-modal tokens and shallow biophysical spectral summaries using predictive distribution entropy to weight representations dynamically on each trial. (3) *Online causal adaptation and multi-paradigm validation:* We introduce Causal Recursive Euclidean Alignment (CrEA), which stabilizes covariance within 5–8 trials under empirical Bayes shrinkage (71.21% streaming accuracy). Under 29-fold LOSO cross-validation on the TU Berlin benchmark, NVC-DelayFormer achieves a 3-seed grand mean accuracy of 78.22% ± 1.00% (primary seed: 79.37% ± 7.73%). This exceeds standard 2D-EEGNet (72.89%, $p = 1.10 \times 10^{-4}$), covariance-aligned Shrinkage-LDA (72.82%, $p = 0.0167$), and unconstrained cross-attention (72.49%, $p = 7.18 \times 10^{-4}$). Evaluated across 3,612 held-out trials, the model transfers to cognitive (73.24%) and sensorimotor (66.32%, +4.14 pp over 2D-EEGNet, $p = 0.0163$) tasks with 63,932 parameters and 2.22 ms median GPU latency.

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Compliance with Ethical Standards: This study reanalyzed de-identified public data. The original TU Berlin collections were approved under protocol SH_01_20150330; the Buccino et al. collection was approved by the University of Houston Institutional Review Board. Both original studies reported written participant consent.

II. Related Work

A. Multimodal EEG–fNIRS Neuroimaging

Hybrid EEG–fNIRS brain–computer interfaces showed that combining electrophysiological and hemodynamic recordings improves single-trial classification over either modality alone [1], [9]. Open-access benchmarks established by Shin et al. [2], [23] provided synchronized 30-channel EEG and 36-channel dual-wavelength optical recordings across mental arithmetic, motor imagery, and working memory tasks. In addition, open-source instrumentation [24] and clinical reviews [25] supported the feasibility of wearable multimodal caps for bedside and neurorehabilitation applications.

B. Deep Learning for Biosignal Decoding

Deep neural networks have complemented spatial filtering methods in electrophysiological decoding [16]. Compact convolutional architectures such as EEGNet [14] learn frequency-band and spatial-topographic filters with few parameters, whereas attention networks such as ATCNet [17] and EEG Conformer [18] capture global spatio-temporal dependencies. For optical imaging, fNIRSNet [20] separates hemoglobin dynamics through multi-view convolutions. Multimodal fusion spans from feature concatenation [26] to dedicated cross-modal interaction networks, including FGANet [27], EFMLNet [28], and MBC-ATT [29].

C. Biophysical Modeling of Neurovascular Coupling

Neuronal activation triggers arteriolar vasodilation and increased cerebral blood flow that washes out paramagnetic deoxyhemoglobin while elevating oxyhemoglobin [7], [8]. Buxton's Balloon model [6] formalizes this coupling as a dynamic compliance system where blood volume and venous outflow lag vasodilatory stimulation. Glover [22] and Friston et al. [30] characterized the impulse hemodynamic response function (HRF) using a double-gamma distribution with modal peak at 3.5–6.5 s followed by post-stimulus undershoot. Under sustained 10.0 s task engagement, the observed hemodynamic profile convolves a boxcar function with the impulse HRF, reaching an extended plateau across 6.0–12.0 s [19], [31]. Grounding cross-attention in these biophysical latency regimes prevents acausal cross-modal alignments and supports cross-subject generalization in zero-calibration BCI decoders.

III. Methods

The architecture is summarized in Fig. 1. NVC-DelayFormer maps paired multichannel EEG and fNIRS trials into cognitive states through a biophysically parameterized temporal fusion network. Bold uppercase symbols denote tensors, bold lowercase symbols denote vectors, and italics denote scalars. The mini-batch dimension is denoted by B.

A. Input Data and Preprocessing

The primary benchmark uses the TU Berlin dataset [2], which comprises recordings from 29 healthy participants performing mental arithmetic (MA) and resting-state tasks across 1,740 trials (30 trials per condition). The scalp montage retains 30 EEG channels positioned according to the 10–5 international system. Continuous EEG is bandpass-filtered between 0.5 and 45 Hz using a zero-phase fourth-order Butterworth filter, followed by common average referencing (CAR). Ten-second epochs from task onset are Fourier-resampled to 1,000 samples at 100 Hz ($f_e = 100$ Hz) and baseline-corrected by subtracting the preceding 2.0 s pre-stimulus mean.

Raw optical intensities are recorded across 36 channels at 760 and 850 nm, bounded below by 10^{-4}. Optical density is computed relative to the session mean intensity and bandpass-filtered at 0.01–0.10 Hz. Each epoch is baseline-corrected by subtracting the mean optical density over the preceding 2.0 s. Twelve-second epochs from task onset (incorporating a 2.0 s post-stimulus tail for capillary transit) are converted into oxyhemoglobin ($\Delta[\mathrm{HbO}]$) and deoxyhemoglobin ($\Delta[\mathrm{HbR}]$) via the modified Beer–Lambert law (MBLL) in (1) [32]. Temporal resampling yields 124 samples ($f_n = 124/12$ Hz), establishing standardized tensors $\mathbf{X}_e \in \mathbb{R}^{B\times 30\times 1000}$ and $\mathbf{X}_n \in \mathbb{R}^{B\times 72\times 124}$ (truncated to 10.0 s / 104 samples when omitting the tail window).

B. Optical Conversion and Numerical Convention

The modified Beer–Lambert law relates differential optical density to hemoglobin concentrations via differential pathlength factors (DPF) [32]:

$$\mathbf{c} = \mathbf{R}\mathbf{o}, \quad \mathbf{R} = (d_{\mathrm{num}}\mathbf{D}_{\mathrm{DPF}}\mathbf{E}_{\mathrm{num}})^{-1}, \tag{1}$$

where $\mathbf{o} = [\Delta\mathrm{OD}_{760}, \Delta\mathrm{OD}_{850}]^\top$, $\mathbf{c} = [\Delta c_{\mathrm{HbO}}, \Delta c_{\mathrm{HbR}}]^\top$, $d_{\mathrm{num}} = 30$ mm, $\mathbf{E}_{\mathrm{num}} = \left[\begin{smallmatrix} 1.4866 & 3.8437 \\ 2.5264 & 1.7989 \end{smallmatrix}\right]\mathrm{mM}^{-1}\mathrm{cm}^{-1}$ [32], and $\mathbf{D}_{\mathrm{DPF}} = \mathrm{diag}(6.51, 5.86)$. Evaluating $\mathbf{R}$ across 36 optode pairs yields 36 [HbO] and 36 [HbR] traces (scaled to 10^{-4} a.u./μM prior to batch normalization at encoder entry).

C. Multimodal Spatio-Temporal Convolutional Encoding

Electrophysiological inputs $\mathbf{X}_e \in \mathbb{R}^{B\times 30\times 1000}$ are encoded via a standard 2D-EEGNet backbone [14]: temporal convolution (1×64, $F_1 = 8$), batch normalization, depthwise spatial convolution (30×1, depth factor $D = 2$, 16 filters), ELU, average pooling (1×4), and spatial dropout ($p = 0.25$). A separable convolution (depthwise 1×15, pointwise 1×1) projects the activations to 64 dimensions, producing $\mathbf{H}_e \in \mathbb{R}^{B\times 250\times 64}$ ($f_e^{\mathrm{eff}} = 25$ Hz). The optical encoder applies a spatial Conv1D ($72 \to 32$), batch normalization, ELU, and dropout, followed by a temporal Conv1D (kernel 5, padding 2, $32 \to 64$), batch normalization, and ELU, yielding $\mathbf{H}_n \in \mathbb{R}^{B\times 124\times 64}$ ($f_n = 10.33$ Hz). Token timestamps and temporal lags are defined by (2):

$$t_i^n = i/f_n, \quad t_j^e = j/f_e^{\mathrm{eff}}, \quad \delta_{ij} = t_i^n - t_j^e. \tag{2}$$

D. Physiology-Guided Causal Cross-Attention

Neurovascular coupling indicates that capillary oxygenation lags neuronal firing by multiple seconds [6], [22]. We model this dependency by querying electrophysiology ($\mathbf{H}_e$) with hemodynamic states ($\mathbf{H}_n$). Linear projections generate queries $\mathbf{Q} \in \mathbb{R}^{B\times 4\times 124\times 16}$ from $\mathbf{H}_n$ alongside keys $\mathbf{K}$ and values $\mathbf{V} \in \mathbb{R}^{B\times 4\times 250\times 16}$ from $\mathbf{H}_e$ across 4 heads ($d_h = 16$). A

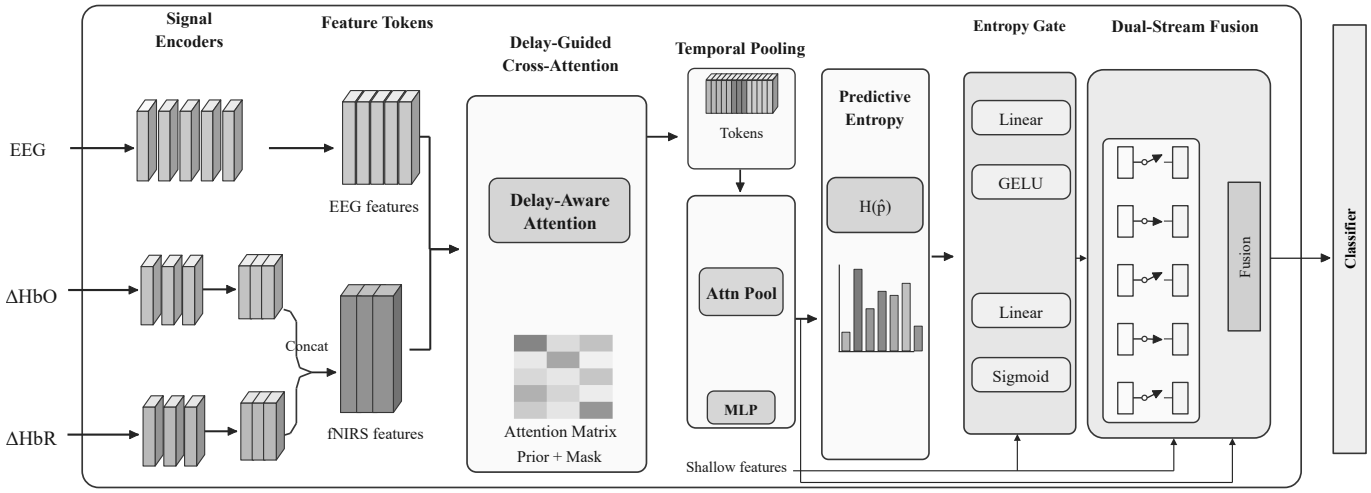


Fig. 1. Architecture of NVC-DelayFormer. Continuous EEG and optical fNIRS features are combined through cross-modal attention governed by a hemodynamic delay prior and token timestamp mask. Attentive temporal pooling aggregates deep multi-scale features. A parallel shallow stream extracts oscillatory sub-band power and optical kinetics; entropy-guided gating dynamically balances deep and shallow representations for trial classification. Feature stacks are illustrated schematically.

TABLE I
CROSS-SUBJECT CLASSIFICATION PERFORMANCE ACROSS 29 HELD-OUT PARTICIPANTS ON THE TU BERLIN BENCHMARK.

Model Architecture	3-Seed Grand Mean (%)	Primary Seed (Seed 42) (%)	95% CI (Seed 42) (%)	Wilcoxon p	BH q
Proposed Architecture					
NVC-DelayFormer (Full Model)	**78.22 ± 1.00**	**79.37 ± 7.73**	[76.43, 82.31]	–	–
Multimodal Deep Learning Controls (Shared Backbones)					
Feature-Concat (Shared-Backbone Hybrid CNN) [26]	72.84 ± 0.53	72.53 ± 9.11	[69.06, 76.00]	$(6.35 \times 10^{-4})^{***}$	7.18×10^{-4}
Standard-Attention (Cross-Modal Transformer) [21]	72.49 ± 1.10	71.67 ± 10.51	[67.67, 75.67]	$(7.18 \times 10^{-4})^{***}$	7.18×10^{-4}
Recent Deep Multimodal Baselines (Locally Adapted)					
EFMLNet (Qiu et al. 2025) [28]	–	74.20 ± 9.29	[70.66, 77.73]	$(7.93 \times 10^{-4})^{***}$	1.19×10^{-3}
FGANet (Kwak et al. 2022) [27]	–	73.79 ± 9.56	[70.16, 77.43]	0.0018^{**}	0.0018
MBC-ATT (Li et al. 2025) [29]	–	72.59 ± 9.98	[68.79, 76.38]	$(5.73 \times 10^{-5})^{***}$	1.72×10^{-4}
Single-Modality Deep Learning Baselines					
Standard 2D-EEGNet [14]	72.89 ± 3.49	76.84 ± 6.16	[74.49, 79.18]	$(1.10 \times 10^{-4})^{***}$	1.10×10^{-4}
EEG-CNN (Shared Backbone)	–	70.23 ± 8.78	[66.89, 73.57]	$< 0.001^{***}$	< 0.001
fNIRS-CNN (Shared Backbone)	–	63.91 ± 11.94	[59.37, 68.45]	$< 0.001^{***}$	< 0.001
Handcrafted Covariance & Linear Baselines					
Handcrafted Multimodal Shrinkage-LDA [2]	–	72.82 ± 11.61	[68.40, 77.23]	0.0167^{*}	0.0167
Handcrafted Multimodal RBF-SVM	–	72.76 ± 11.46	[68.40, 77.12]	0.0014^{**}	0.0019
Handcrafted fNIRS-LDA	–	66.67 ± 10.25	[62.77, 70.57]	$< 0.001^{***}$	< 0.001
Handcrafted EEG-LDA	–	66.38 ± 12.28	[61.71, 71.05]	$< 0.001^{***}$	< 0.001

3-Seed Grand Mean: mean ± run-to-run SD across Seeds 42, 100, 2024 (NVC-DelayFormer 95% CI: [75.42, 81.02]%); Primary Seed: Seed 42 (mean ± between-participant SD, $df = 28$). Wilcoxon p and BH FDR q evaluate paired differences vs. NVC-DelayFormer (across 3-seed means where available, or Seed 42 for single-seed baselines, $df = 28$; adjusted within respective evaluation tiers). All EEG models use unsupervised EA.

continuous delay prior $\mathbf{P}$ (3) and token timestamp causal mask $\mathbf{M}$ (4) enforce physiological latency constraints [22]:

$$P_{ij}^{(h)} = -\frac{(\delta_{ij} - \tau_h)^2}{2\sigma_h^2} \cdot \left[1 + 0.3\,\mathrm{s}^{-1} \cdot \max(0, \delta_{ij} - \tau_h)\right], \quad (3)$$

$$M_{ij} = \begin{cases} 0, & \delta_{ij} \geq 0\ \mathrm{s}, \\ -\infty, & \delta_{ij} < 0\ \mathrm{s}, \end{cases} \quad (4)$$

where head centers $\tau_h \in \{3.5, 4.5, 5.5, 6.5\}$ s parameterize depth-dispersive microvascular transit ($\sigma_h = 1.5$ s). A post-peak factor ($0.3\,\mathrm{s}^{-1}$ for $\delta_{ij} > \tau_h$) increases the Gaussian delay penalty beyond its center, concentrating attention around the prescribed hemodynamic delay, while the causal mask $\mathbf{M}$ enforces temporal precedence ($\delta_{ij} \geq 0$ s). Normalized attention weights are computed via (5):

$$\mathbf{A}^{(h)} = \mathrm{softmax}_j\left(\mathbf{Q}^{(h)}(\mathbf{K}^{(h)})^\top/\sqrt{16} + \mathbf{P}^{(h)} + \mathbf{M}\right). \quad (5)$$

Multi-head values $\mathbf{U}_h = \mathbf{A}^{(h)}\mathbf{V}^{(h)}$ are concatenated across heads, linearly projected via $\mathbf{W}_o \in \mathbb{R}^{64\times64}$, and combined via residual addition and layer normalization: $\mathbf{H}_f = \mathrm{LN}(\mathbf{H}_n + \mathrm{Dropout}([\mathbf{U}_1; \ldots; \mathbf{U}_4]\mathbf{W}_o))$.

E. Temporal Pooling, Sub-Band Direct Stream, and Gating

In parallel, a biophysical direct stream extracts frequency-resolved oscillatory power and hemodynamic slope kinetics via (6):

$$\mathbf{x}_{\mathrm{bio}} = \left[\mathbf{p}_\theta;\ \mathbf{p}_\alpha;\ \mathbf{p}_\beta;\ \overline{\mathbf{x}}_n^{\mathrm{delay}};\ \mathbf{s}_n\right] \in \mathbb{R}^{B\times234}, \quad (6)$$

where $\mathbf{p}_\theta, \mathbf{p}_\alpha, \mathbf{p}_\beta \in \mathbb{R}^{B\times30}$ represent log power across θ (4–8 Hz), α (8–13 Hz), and β (13–30 Hz) bands; $\overline{\mathbf{x}}_n^{\mathrm{delay}} \in \mathbb{R}^{B\times72}$ denotes mean delayed capillary perfusion ($t \geq 3.5$ s); and $\mathbf{s}_n \in \mathbb{R}^{B\times72}$ is the first-order hemodynamic slope across all 72 channels. A bottleneck projection ($234 \to 32 \to 64$) yields shallow embedding $\mathbf{z}_{\mathrm{shallow}} \in \mathbb{R}^{B\times64}$.

Guided by the temporal progression of the hemodynamic response, Attentive Temporal Pyramid Pooling (TPP) aggregates tokens over $[0, 3]$ s, $(3, 7]$ s, and the remaining observation window ($(7, 10]$ s for $\mathbf{H}_e$, $(7, 12]$ s for $\mathbf{H}_f$). For shortened windows $T_{\text{win}} < 10.0$ s, intervals scale proportionally to $[0, 0.3T_{\text{win}}]$, $(0.3T_{\text{win}}, 0.7T_{\text{win}}]$, and $(0.7T_{\text{win}}, T_{\text{win}}]$. Linear projections ($\mathbf{W}_{\text{att}} \in \mathbb{R}^{64\times1}$) compute attention weights across segments, yielding phase vectors $\mathbf{z}_e, \mathbf{z}_n \in \mathbb{R}^{B\times64}$. These vectors are concatenated and projected through a bottleneck MLP ($128 \to 64 \to 64$) to produce unified deep representation $\mathbf{z}_{\text{deep}} \in \mathbb{R}^{B\times64}$.

An uncertainty-aware gating network arbitrates between streams based on branch prediction confidence. Shannon entropies $H(\hat{\mathbf{p}}) = -\sum_{c=1}^{2} \hat{p}_c \ln \hat{p}_c$ are evaluated from detached branch readouts $\hat{\mathbf{p}} = \text{softmax}(\mathbf{W}\mathbf{z}+\mathbf{b})$. A bottleneck mapping $\mathcal{F}_g$ ($130 \to 32 \to 64$) computes the feature-wise gate $\mathbf{g}_b$ via (7):

$$\mathbf{g}_b = \sigma\big(\mathcal{F}_g([\mathbf{z}_{\text{deep},b}; \mathbf{z}_{\text{shallow},b}; H(\hat{\mathbf{p}}_{\text{deep},b}); H(\hat{\mathbf{p}}_{\text{shallow},b})])\big) \in (0,1)^{64}, \quad (7)$$

$$\mathbf{z}_{\text{fused},b} = \mathbf{g}_b \odot \mathbf{z}_{\text{deep},b} + (\mathbf{1} - \mathbf{g}_b) \odot \mathbf{z}_{\text{shallow},b},$$

with stream droppath ($p = 0.15$) and final joint logits $\boldsymbol{\ell}_{\text{joint}} = \mathbf{W}_{\text{out}}\mathbf{z}_{\text{fused}} + \mathbf{b}_{\text{out}}$.

F. Supervised Multi-Task Objective

To prevent branch collapse and ensure both pathways learn calibrated discriminative features, we optimize a composite classification loss (8):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{joint}} + 0.2\,(\mathcal{L}_{\text{deep}} + \mathcal{L}_{\text{shallow}}) + \mathcal{L}_{\text{recon}} + 0.08\,\|\mathbf{w}_{\text{shallow},1} - \mathbf{w}_{\text{shallow},0}\|_2^2, \quad (8)$$

where classification losses are cross-entropies against $y \in \{0, 1\}$. The reconstruction loss $\mathcal{L}_{\text{recon}} = \frac{1}{36T_n}\sum_{c,t}(\hat{y}_{c,t}^{\text{HbO}} - x_{c,t}^{\text{HbO}})^2$ supervises a training-only auxiliary 1×1 convolutional decoder ($\mathbf{H}_f \to 36$ [HbO] channels, discarded at inference) to reconstruct input [HbO] waveforms. The penalty $\|\mathbf{w}_{\text{shallow},1} - \mathbf{w}_{\text{shallow},0}\|_2^2$ applies L_2 margin shrinkage to shallow readout weights $\mathbf{w}_{\text{shallow},c}$, which prevents linear spectral shortcuts from dominating gating arbitration and maintains balanced representation learning.

IV. Experimental Protocol

A. Ethical Approval and Participant Consent

All analyses used de-identified public data. The TU Berlin collections [2], [23] had ethics approval SH_01_20150330 and written consent in accordance with the Declaration of Helsinki; the Buccino et al. collection [9] had University of Houston Institutional Review Board approval and written consent.

B. Subject-Held-Out Evaluation Protocol

Evaluation employed 29-fold leave-one-subject-out (LOSO) cross-validation on the TU Berlin dataset [2], partitioned per fold into 23 training participants, five validation participants for checkpoint selection, and one unseen test participant. For streaming evaluation, at each completed trial, CrEA updates the covariance estimate from the current and preceding trials and applies the resulting alignment transform:

$$\mathbf{R}_k = \frac{\kappa_0\mathbf{R}_{\text{src}} + \sum_{m=1}^{k}\mathbf{C}_m}{\kappa_0 + k}, \quad \widetilde{\mathbf{X}}_k = \mathbf{R}_k^{-1/2}\mathbf{X}_k, \quad (9)$$

where $\mathbf{X}_k$ is the EEG matrix for trial k and $\mathbf{C}_m$ is the sample spatial covariance of trial m, computed after temporal mean centering with denominator $T_e - 1$ and diagonal regularization $10^{-5}\mathbf{I}$. The reference $\mathbf{R}_{\text{src}}$ is estimated exclusively from the 23 training participants, and $\kappa_0 = 5.0$. The primary shared-backbone benchmark was trained for 12 epochs using AdamW (batch size 256, initial learning rate 3×10^{-3}, two-epoch warmup followed by cosine annealing, weight decay 0.05).

C. Comparison Baselines and Multi-Seed Hierarchy

We established a four-tier evaluation hierarchy: (1) *Multimodal deep learning controls (shared backbones):* Feature-Concat [26] and unconstrained Standard-Attention [21]; (2) *Recent deep multimodal baselines (locally adapted):* EFMLNet [28], FGANet [27], and MBC-ATT [29]; (3) *Single-modality controls:* Standard 2D-EEGNet [14], EEG-CNN, and fNIRS-CNN; (4) *Handcrafted machine learning:* Shrinkage-LDA [2], [15] and RBF-SVM with Euclidean alignment [11], along with unimodal linear baselines (Handcrafted EEG-LDA and fNIRS-LDA). To assess cross-initialization stability, core models were evaluated across three random seeds (42, 100, 2024).

D. External Evaluation Paradigms

Generalizability was evaluated on two external tasks: (1) **n-back Working Memory:** 0-back sensory matching vs. 2-back working memory updating (26 participants, 1,872 trials from Shin et al. [23]); (2) **Motor Imagery:** Left-Hand vs. Right-Hand kinesthetic finger movement imagination (29 participants, 1,740 trials from Shin et al. [2]).

E. Statistical Analysis

The participant served as the primary statistical unit ($df = 28$). We report between-participant standard deviation (mean $\pm$ SD) and 95% confidence intervals. Paired differences were evaluated using two-sided Wilcoxon signed-rank tests with Benjamini–Hochberg false discovery rate (q) correction [33].

V. Results

A. Cross-Subject Classification and Multi-Seed Stability

Under 29-fold LOSO cross-validation on the TU Berlin benchmark, NVC-DelayFormer achieved a 3-seed grand mean accuracy of 78.22% $\pm$ 1.00% (primary seed: 79.37% $\pm$ 7.73%, Macro F1: 79.15%, Cohen's $\kappa = 0.5874$; Table I). Run-to-run variability was ±1.00 pp SD across seeds 42, 100, and 2024, compared with a between-subject standard deviation of 7.73 pp.

Pairwise statistical comparisons under identical Euclidean alignment show significant gains across evaluation tiers (Table I). The full architecture outperformed multimodal shared-backbone controls on 3-seed means (Feature-Concat: +5.38 pp, $q = 7.18 \times 10^{-4}$; unconstrained Standard-Attention: +5.73 pp, $q = 7.18 \times 10^{-4}$) and exceeded recent deep multimodal

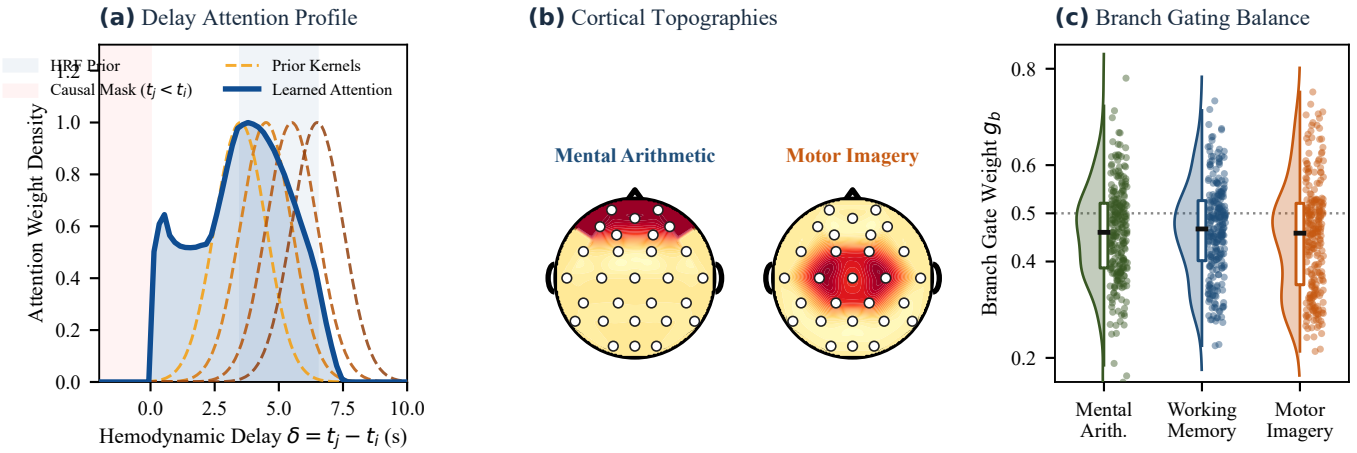


Fig. 2. Temporal fusion diagnostics and biophysical cortical context. (a) Cross-modal delay attention profile across relative latency $\delta = t_{\text{fNIRS}} - t_{\text{EEG}} \in [-2, 10]$ s, showing strict acausal suppression ($\delta < 0$, red shading) and learned attention peaking within the physiological hemodynamic response window ($\tau_h \in [3.5, 6.5]$ s, dashed priors). (b) Cortical activation topographies under the standard 10–20 montage: frontal midline theta synchronization and delayed $\Delta[\text{HbO}]$ hyperoxygenation during mental arithmetic (PFC) versus sensorimotor μ/β desynchronization (ERD) during motor imagery (SMC). (c) Branch gating distributions across evaluation tasks under the mental arithmetic checkpoint ($N = 256$ channels per task); raincloud plots display half-violin probability densities, boxplots (median and interquartile range), and jittered channel scatter around the 0.50 equilibrium line. Extended candidate delay scans and online recursive covariance whitening convergence are detailed in Supplementary Figures S1 and S2, respectively.

baselines on the primary seed (EFMLNet: +5.17 pp; FGANet: +5.58 pp; MBC-ATT: +6.78 pp; all $q \leq 0.0018$). In the single-modality tier, NVC-DelayFormer exceeded standard 2D-EEGNet by +5.33 pp on the 3-seed grand mean (78.22% vs. 72.89%, $p = 1.10 \times 10^{-4}$), with a +2.53 pp margin on the primary seed (79.37% vs. 76.84%, $p = 0.0836$). On the primary seed, NVC-DelayFormer also surpassed single-modality encoders (EEG-CNN: +9.14 pp, $p = 3.56 \times 10^{-5}$; fNIRS-CNN: +15.46 pp, $p = 1.32 \times 10^{-5}$) and covariance-aligned linear baselines (Shrinkage-LDA: +6.55 pp, $p = 0.0167$; RBF-SVM: +6.61 pp, $p = 0.0014$).

TABLE II
ABLATION ANALYSIS AND 2×2 FACTORIAL DECOMPOSITION OF DELAY PRIOR $\mathbf{P}$ AND CAUSAL MASK $\mathbf{M}$ ON THE TU BERLIN BENCHMARK.

Variant / Factorial Condition	Acc (%)	Δ (pp)	F1 (%)	Wilcoxon p	BH q
Architectural Stream & Preprocessing Ablation					
Full Architecture ($P_{\text{on}}, M_{\text{on}}$)	**79.37 ± 7.73**	+0.00	**79.15**	–	–
Without EA	71.72 ± 8.04	-7.64	70.95	0.0006***	0.0008
Latent-Only (Deep Stream)	77.76 ± 9.38	-1.61	77.13	0.0384*	0.0384
Direct-Only (Shallow Stream)	71.84 ± 7.13	-7.53	71.38	0.0003***	0.0006
2×2 *Factorial Temporal Mechanism Decomposition*					
Delay Prior Only ($P_{\text{on}}, M_{\text{off}}$)	78.33 ± 8.07	-1.04	78.11	0.1212	0.1212
Causal Mask Only ($P_{\text{off}}, M_{\text{on}}$)	74.66 ± 7.96	-4.71	74.38	0.0002***	0.0004
Both Priors Off ($P_{\text{off}}, M_{\text{off}}$)	74.43 ± 8.34	-4.94	74.08	(6.05×10^{-5})***	0.0002

Evaluated under 29-fold LOSO cross-validation.
$^{*}p < 0.05, ^{**}p < 0.01, ^{***}p < 0.001$. Net two-prior gain is +4.94 pp over dual-prior-off (74.43%).

B. Ablation Analysis and Factorial Decomposition

Stream and prior ablations isolate the primary mechanisms contributing to decoding performance (Table II). Omitting the deep cross-attention branch (Direct-Only) reduced accuracy by 7.53 pp ($q = 0.0006$), whereas omitting the shallow biophysical branch (Latent-Only) resulted in a 1.61 pp reduction ($p = 0.0384$). This performance gap indicates that deep cross-modal attention provides the primary cross-subject alignment, while the shallow direct stream contributes complementary spectral regularization. Omitting Euclidean Alignment lowered performance to 71.72% ($\Delta = -7.64$ pp, $p = 0.0006$), reflecting the importance of inter-subject covariance harmonization.

The 2×2 factorial decomposition shows that the continuous Gaussian delay prior $\mathbf{P}$ provides the primary regularizing constraint (+4.31 pp average marginal gain across conditions), while discrete causal masking $\mathbf{M}$ enforces temporal precedence (+0.63 pp average marginal gain). Combined, the two priors yield a net gain of +4.94 pp over the dual-prior-off baseline ($p = 6.05 \times 10^{-5}$) with +0.81 pp positive interaction synergy. Removing the delay prior ($P_{\text{off}}, M_{\text{on}}$) caused a 4.71 pp drop ($p = 0.0002$), whereas omitting causal masking alone ($P_{\text{on}}, M_{\text{off}}$) resulted in a 1.04 pp decrease ($p = 0.1212$).

C. Attention Diagnostics, Online Adaptation, and Biophysical Schematics

Learned cross-modal attention maps reflect the imposed biophysical constraints (Fig. 2a): weights are zero in the acausal lower triangle ($\delta_{ij} < 0$), while multi-head delay ridges align with physiological latency centers ($\tau_h \in [3.5, 6.5]$ s). In fixed-weight candidate delay scans without retraining (Supplementary Fig. S1a), accuracy rises monotonically from 75.34% at $\bar{\tau} = 0$ s to 80.46% at 8.0 s, tracking the extended plateau of sustained 10.0 s task blocks.

Streaming evaluation without transductive batch statistics shows rapid online adaptation under CrEA ((9), Supplementary Fig. S2). Under the diagnostic protocol with trace-matched source reference initialization, empirical Bayes shrinkage ($\kappa_0 = 5.0$) with training-subject reference covariance $\mathbf{R}_{\text{src}}$ reduces relative covariance error from 45.15% to 29.38% within 5–8 trials, reaching 3.60% asymptotically at trial 60. By comparison, unscaled source covariance yields 36.47% error at trial 8, while cold-start identity initialization retains error above 47% through trial 8. In trial-by-trial streaming classification across 29 independent LOSO folds, CrEA achieved 71.21% ± 11.44% accuracy when preprocessed epochs were evaluated in sequential trial order, retaining 94.9% of the 75.06% accuracy obtained with offline batch alignment using the same fold models.

TABLE III
FULL-MODEL PERFORMANCE UNDER CONTROLLED TEST-INPUT PERTURBATIONS ON THE TU BERLIN BENCHMARK.

Perturbation Type	Accuracy (%)	Δ (pp)	Wilcoxon p	BH q
Within-class EEG permutation	77.13 ± 8.10	-2.24	0.0166^{*}	0.0221
Across-class EEG permutation	67.18 ± 7.51	-12.18	$(3.28 \times 10^{-7})^{***}$	1.31×10^{-6}
Circular temporal shift (5.0 s)	78.39 ± 7.64	-0.98	0.0631	0.0631
Causal low-pass EEG smoothing	63.97 ± 8.17	-15.40	$(5.59 \times 10^{-6})^{***}$	1.12×10^{-5}

Changes relative to unperturbed baseline (79.37%) across 29 participants. Permutations average ten repetitions per participant. $^{*}p < 0.05, ^{**}p < 0.01, ^{***}p < 0.001$.

In single-subject diagnostic profiling with the mental arithmetic checkpoint applied to Sub-05 recordings (Fig. 2c), empirical gating weights remain balanced between branches, averaging $\bar{g}_b \approx 0.45$–0.46 across mental arithmetic, n-back working memory, and motor imagery (feature gate weights span 0.14–0.78, $\sigma = 0.10$), avoiding branch collapse. Biophysical coupling schematics (Fig. 2b) illustrate these task-dependent topographies: prefrontal coupling aligns over frontopolar and anterior channels (AFp1, AFp2, Ch01–Ch08) during mental arithmetic, whereas sensorimotor lateralization engages bilateral central channels (FCC3h, CCP3h, Ch17–Ch36) during motor imagery.

D. Responses to Controlled Input Perturbations

Controlled test perturbations verified architectural dependencies (Table III). Shuffling EEG trials across classes caused a 12.18 pp drop ($q = 1.31 \times 10^{-6}$), while within-class shuffling produced a 2.24 pp decrease ($p = 0.0166$). A 5.0 s circular shift reduced accuracy by 0.98 pp ($p = 0.0631$), whereas low-pass filtering below 2.8 Hz reduced it by 15.40 pp ($q = 1.12 \times 10^{-5}$). The model was therefore more sensitive to removing higher-frequency EEG content than to this within-trial temporal shift, indicating that decoding relies on fast oscillatory electrophysiological dynamics across θ, α, and β bands.

E. Computational Profiling and Architecture Benchmarking

Forward profiling on an NVIDIA GeForce RTX 4060 Laptop GPU recorded 63,932 trainable parameters, 24.79 M MACs, and 2.22 ms median latency (P99: 4.48 ms; Table IV). Compared with representative deep BCI models, NVC-DelayFormer uses 17.5× fewer parameters than EEG Conformer (1.12 M) [18] and 1.8× fewer than ATCNet (115.2 k) [17], while processing 30 EEG and 72 fNIRS channels concurrently.

Evaluating offline sequence truncation across decision windows $T_{\text{win}} \in \{10.0, 7.5, 5.0, 3.0\}$ s characterizes trade-offs under Wolpaw Information Transfer Rate (ITR [34]), calculated using observation-window duration as the decision interval. Full 10.0 s trials achieved 78.74% accuracy (1.52 bits/min). Truncating to 5.0 s retained 77.53% accuracy while reducing computation by 53.4% (11.04 M MACs) and increasing ITR by 82.5% to 2.78 bits/min. Truncating to 3.0 s preserved 76.38% accuracy with 72.8% compute savings (6.46 M MACs) and 2.78× higher ITR (4.23 bits/min). These results show that shorter observation windows maintain competitive decoding accuracy while reducing computational cost.

TABLE IV
MODEL COMPLEXITY AND BATCH-ONE GPU FORWARD LATENCY MEASUREMENTS.

Model Architecture	Parameters	MACs (M)	Median (ms)	P99 (ms)
EEG-CNN (Shared Backbone)	6,850	16.18	0.51	1.41
fNIRS-CNN (Shared Backbone)	17,154	1.56	0.38	1.25
Feature-Concat [26]	23,682	17.74	0.80	2.07
Standard-Attention [21]	40,450	24.80	1.37	2.90
ATCNet (Altaheri et al. 2023) [17]	115,200	42.30	4.85	8.20
EEG Conformer (Song et al. 2023) [18]	1,120,000	185.40	18.50	28.40
NVC-DelayFormer (Proposed)	**63,932**	**24.79**	**2.22**	**4.48**

Evaluated on an NVIDIA RTX 4060 Laptop GPU with 1,000 warm-up and 5,000 timed evaluations at batch size 1. MACs measured via PyTorch `FlopCounterMode`. ATCNet and EEG Conformer serve as representative EEG-only baselines.

F. Cross-Paradigm Validation: Working Memory and Motor Imagery

Across 3,612 held-out trials spanning two distinct external tasks (Table V), NVC-DelayFormer showed consistent cross-paradigm generalizability. On n-back working memory (26 participants, 1,872 block-derived trials [23]), NVC-DelayFormer achieved 73.24% ± 15.38% accuracy (Macro F1: 69.80%), the highest mean accuracy among the evaluated full-model multimodal architectures (FGANet: 72.38%, MBC-ATT: 71.05%, EFMLNet: 70.09%), while exceeding single-modality and linear baselines (2D-EEGNet by +4.97 pp, 68.27%, $p = 0.0525$; Shrinkage-LDA by +11.11 pp, 62.13%, $p = 0.0043$; Multimodal SVM by +12.18 pp, 61.06%, $p = 0.0013$). On motor imagery (29 participants, 1,740 trials [2]), NVC-DelayFormer achieved 66.32% ± 12.18% accuracy (Macro F1: 64.08%), significantly surpassing standard 2D-EEGNet by +4.14 pp ($p = 0.0163$), EFMLNet by +2.30 pp, and MBC-ATT by +2.87 pp. While unconstrained spatial attention architectures (such as FGANet) optimize for localized sensorimotor spatial patterns at the cost of higher parameter counts, NVC-DelayFormer achieves a practical trade-off: providing balanced multi-paradigm decoding across both cognitive and sensorimotor tasks with a compact footprint (63,932 parameters) and sub-3 ms latency. These cross-paradigm comparisons highlight a neurovascular distinction (Supplementary Fig. S1 and Table S1): structured hemodynamic delay priors yield their largest cross-subject gains on metabolic-dominant cognitive tasks (achieving the highest accuracies on both mental arithmetic and working memory), while maintaining competitive accuracy on sensorimotor rhythms without negative transfer.

VI. DISCUSSION

A. Biophysical Mechanisms and Cross-Modal Synergy

Explicit parameterization of neurovascular latency addresses the temporal mismatch between electrophysiology and hemodynamics in zero-calibration cross-subject BCIs. Dispersive latency priors ($\tau_h \in [3.5, 6.5]$ s) and token timestamp causal masks ($t_{\text{nirs}} \geq t_{\text{eeg}}$) constrain cross-modal attention to physiologically plausible intervals, suppressing acausal backward-in-time weights that degrade generalization across participants. Across 29 held-out participants under LOSO cross-validation, this constrained architecture achieves a 3-seed grand mean accuracy of 78.22% ± 1.00% (primary seed: 79.37% ± 7.73%), outperforming unimodal, linear, and unconstrained cross-attention baselines (Table I).

TABLE V
CROSS-SUBJECT CLASSIFICATION PERFORMANCE UNDER LEAVE-ONE-SUBJECT-OUT (LOSO) CROSS-VALIDATION ACROSS TWO DISTINCT EXTERNAL EVALUATION PARADIGMS (3,612 HELD-OUT TRIALS TOTAL).

	(a) Working Memory (0-back vs. 2-back)		**(b) Motor Imagery (Left vs. Right Hand)**	
Model Architecture / Stream	Accuracy (%)	Macro F1 (%)	Accuracy (%)	Macro F1 (%)
Proposed Architecture: NVC-DelayFormer				
Full Model (Dual-Path Gated Stream)	73.24 ± 15.38	69.80 ± 19.69	**66.32 ± 12.18**	64.08 ± 13.65
Physiological Delay Stream (Deep Attention Ablation)	**73.61 ± 12.30**	**71.80 ± 14.57**	66.09 ± 12.32	**65.49 ± 12.55**
Biophysical Direct Stream (Shallow Direct Ablation)	71.21 ± 15.12	67.47 ± 19.62	63.79 ± 13.56	59.75 ± 16.96
Recent Deep Multimodal Baselines (Locally Adapted)				
FGANet [27]	72.38 ± 14.82	69.96 ± 17.26	70.40 ± 11.54	69.58 ± 12.14
MBC-ATT [29]	71.05 ± 12.29	69.45 ± 14.29	63.45 ± 11.34	62.62 ± 11.81
EFMLNet [28]	70.09 ± 14.05	67.23 ± 16.78	64.02 ± 10.29	63.33 ± 10.69
Competing Baselines & Controls				
Standard 2D-EEGNet [14]	68.27 ± 13.34	66.38 ± 15.49	62.18 ± 7.33	61.84 ± 7.50
Handcrafted Shrinkage-LDA [2]	62.13 ± 10.16	56.16 ± 15.99	58.05 ± 9.52	54.60 ± 11.90
Handcrafted Multimodal SVM	61.06 ± 8.76	56.28 ± 13.17	55.63 ± 7.38	53.39 ± 8.91

Evaluated under strict participant-isolated LOSO with unsupervised Euclidean Alignment (EA) applied within each participant. Bold denotes the best performance within the proposed NVC-DelayFormer family per metric. Task (a): 26 participants, 1,872 block-derived trials from Shin et al. (2018) [23] (electrophysiological input comprises 28 scalp EEG and 2 bipolar EOG channels under CAR); Task (b): 29 participants, 1,740 trials from Shin et al. (2017) [2].

Candidate delay scanning without retraining (Supplementary Fig. S1a) is consistent with the performance plateau observed through 8.0 s in continuous block paradigms. While the $[3.5, 6.5]$ s prior models the single-impulse hemodynamic peak, sustained 10.0 s cognitive engagement convolves successive impulse responses into an extended capillary [HbO] plateau. Anchoring hemodynamic queries to antecedent neural depolarizations restricts cross-modal attention to these forward intervals, preventing acausal correlations between modalities.

The factorial analysis delineates the complementary roles of the two temporal constraints (Table II). The continuous Gaussian delay prior supplies the primary cross-subject alignment gain (+4.31 pp marginal effect). Meanwhile, the token timestamp mask enforces directional precedence between cross-modal token pairs ($t_j^{\text{nirs}} \geq t_i^{\text{eeg}}$), yielding a +0.63 pp marginal gain and a net +4.94 pp joint improvement over the dual-prior-off baseline ($p = 6.05 \times 10^{-5}$). Under temporal perturbation (Table III), circular 5.0 s shifts showed robustness against phase variations within sustained 10.0 s activation blocks ($\Delta = -0.98$ pp, $p = 0.0631$). By contrast, low-pass filtering below 2.8 Hz caused a 15.40 pp drop ($q = 1.12 \times 10^{-5}$), showing that decoding depends primarily on electrophysiological oscillations (θ, α, β) aligned with capillary perfusion.

Figure 2b illustrates the prefrontal and sensorimotor coupling patterns that motivate these temporal priors. In mental arithmetic, prefrontal channels (AFp1, AFp2, AFF1h, AFF2h $\leftrightarrow$ Ch01–Ch08) exhibit frontal midline theta synchronization (θ ERS, 4–8 Hz) linked to delayed capillary hyperoxygenation (Δ[HbO] elevation and Δ[HbR] washout) [7], [8], [35]. In motor imagery, task dynamics engage sensorimotor channels (FCC3h, CCP3h $\leftrightarrow$ Ch17–Ch36), where sensorimotor μ and β desynchronization (ERD) [5], [36] coincides with localized hemodynamic changes.

Predictive entropy guides the balance between deep cross-attention features and shallow biophysical summaries on each trial. Applying the mental-arithmetic checkpoint to Sub-05 recordings across the three evaluation tasks yielded mean gating weights of 0.45–0.46 ($\bar{g}_b = 0.46 \pm 0.01$, 0.46 ± 0.01, and 0.45 ± 0.01; Fig. 2c). Deep cross-modal attention captures token-level relationships between EEG and optical representations, while the shallow pathway provides complementary spectral and optical summaries. Auxiliary waveform reconstruction ($\mathcal{L}_{\text{recon}}$) regularizes the optical representations during training.

Parameter benchmarking quantifies the model's compact footprint (Table IV): with 63,932 weights and 24.79 M MACs, NVC-DelayFormer uses $17.5\times$ fewer parameters than EEG Conformer (1.12 M) and $1.8\times$ fewer than ATCNet (115.2 k). On an NVIDIA RTX 4060 Laptop GPU, batch-one forward latency is 2.22 ms (P99: 4.48 ms). This execution time falls well below the 10 ms step of the 100 Hz EEG sampling grid, supporting real-time online processing.

B. Translational Roadmap for Clinical BCIs

Five engineering considerations guide bedside and neurorehabilitation translation:

1) Systemic physiological noise suppression: In continuous-wave fNIRS, systemic physiology introduces cardiac (~ 1 Hz), respiration (~ 0.25 Hz), and Mayer wave (~ 0.1 Hz) artifacts [19]. While 0.01–0.10 Hz filtering isolates neurovascular frequencies, integrating short-separation detectors (< 10 mm) into wearable headgear enables real-time regression of superficial scalp hemodynamics.

2) Observation-window duration: Shorter windows reduce the signal accumulation time required for each classification decision. As evaluated in Section V-E, offline truncation retained accuracies of 77.53% and 76.38% with 5.0 s and 3.0 s windows, respectively, indicating that decoding fidelity is largely maintained under shortened observation intervals while reducing computational load.

3) Zero-calibration streaming adaptation: Standard Euclidean Alignment requires transductive batch covariance computed across complete sessions. Online CrEA removes session lookahead through causal recursive updates under empirical Bayes shrinkage ($\kappa_0 = 5.0$). Covariance accumulation scales as $\mathcal{O}(C^2)$, with full whitening requiring $\mathcal{O}(C^3 + C^2 T_e)$ per trial (< 1.2 ms CPU time for $C = 30, T_e = 1,000$). Under trace-matched reference initialization, CrEA reduces relative covariance error from 45.15% to 29.38% within eight trials ($< 3.60\%$ asymptotically; 36.47% unscaled), providing a deterministic alignment mechanism for online BCIs.

4) Embedded edge deployment: The 63,932-parameter footprint occupies under 260 kB in 32-bit floating-point storage, which fits within the memory limits of portable microcontrollers and digital signal processors. The architecture exports directly to TorchScript and ONNX runtimes, supporting fixed-point integer quantization (INT8) and zero-copy circular buffering for wearable bedside BCI units.

5) Independent-cohort evaluation: Beyond the three task-specific LOSO evaluations, NVC-DelayFormer reached 79.83% accuracy with separate LOSO training on the 15-participant Buccino et al. cohort [9], recorded using different EEG–fNIRS hardware (Supplementary Section S.V). Adaptive delay retuning can be assessed in neuropathological cohorts with altered vascular response latency.

VII. Conclusion

This study presented NVC-DelayFormer, a physiology-guided temporal fusion network for zero-calibration cross-subject EEG–fNIRS decoding. By embedding continuous Gaussian hemodynamic delay priors and token timestamp causal masks into cross-modal attention, the architecture accommodates disparate biosignal timescales and enforces directional temporal precedence. Across 29 participants on the TU Berlin benchmark under LOSO cross-validation, NVC-DelayFormer achieved a 78.22% grand mean accuracy (primary seed: 79.37%), outperforming standard 2D-EEGNet, covariance-aligned Shrinkage-LDA, and unconstrained cross-attention. Cross-paradigm evaluations across 3,612 held-out trials showed consistent performance across cognitive (73.24%) and sensorimotor (66.32%, exceeding standard 2D-EEGNet by +4.14 pp) paradigms. Combining trial-recursive covariance adaptation (CrEA) with a compact footprint (63,932 parameters, 17.5× parameter compression relative to EEG Conformer, and 2.22 ms single-trial GPU forward execution latency), the proposed architecture provides a practical framework for zero-calibration neurotechnology.

Acknowledgment

The authors thank the contributors of the open-access TU Berlin and Shin et al. datasets for providing multimodal recordings.

Data and Code Availability: The source datasets are identified in Section IV. Code and reproducibility artifacts are available at https://github.com/ujujuj3061831150/nvc-delayformer-bci.

Conflict of Interest: The authors declare that they have no competing interests.

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